Ericsson PE Network SOP for QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) on Node WHI-PE-14 at Whitefield East POP

Node: WHI-PE-14 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_7848

Description

This SOP addresses the critical QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) failure on the Ericsson PE Node WHI-PE-14 at Whitefield East POP. This failure domain occurs due to the inherent limitations of the PE role in routing table overflow, traffic congestion, or QoS misconfiguration. In the Ericsson equipment, the PE role utilizes the QoS framework to prioritize and manage traffic across the network. However, if the QoS configuration becomes misaligned with the network's traffic patterns, or if the network experiences a sudden surge in traffic, it can lead to QoS queue starvation, microbursts, or interface tail drops. These events can result in network congestion, packet loss, and ultimately, the ALARM_7848 Critical alarm. To prevent such failures, it is crucial to regularly monitor QoS configurations, traffic patterns, and network performance to ensure optimal operation. Furthermore, this SOP outlines the steps necessary to diagnose and resolve the critical QoS / Congestion Event on the Ericsson PE Node WHI-PE-14.

Pre-requisite Checks

- check: Verify the status of the redundant link to WHI-PE-14 at Whitefield East POP., action: Dispatch remote hands to check the link status and ensure redundancy.

- check: Check the current QoS configuration on WHI-PE-14., action: Execute the command 'show qos configuration' to verify QoS settings.

- check: Verify the network traffic patterns and QoS statistics., action: Execute the command 'show qos statistics' to analyze traffic data.

Potential Root Causes

Cause	Probability
Misconfigured QoS settings	High
Network congestion due to traffic surge	Medium
QoS queue starvation	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos configuration | include policy-map, Expected Output:
Policy-map name: PE-traffic-policy, Description: This command displays
```

```
the QoS policy configuration on WHI-PE-14, focusing on the policy-map
name, which is crucial for identifying potential misconfigurations.

Command: show interface GigabitEthernet 0/0/0 | include queue,
Expected Output: Queueing strategy: priority-queue, Description: This
command displays the interface queue configuration, providing insight
into the queueing strategy and potential tail drops.

Command: show qos statistics | include packets, Expected Output:
Packets transmitted: 1234567, Description: This command displays QoS
statistics, including packets transmitted, which can help identify
potential QoS queue starvation.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS configuration, Command: config qos policy-map PE-traffic-policy, Description: Verify the QoS policy configuration and adjust it as necessary to ensure optimal traffic prioritization and management.

Step 2: Step: Review and adjust interface queue configuration, Command: config interface GigabitEthernet 0/0/0 queue strategy priority-queue, Description: Review the interface queue configuration and adjust it as necessary to prevent tail drops and ensure optimal traffic transmission.

Step 3: Step: Monitor and adjust network traffic patterns, Command: show qos statistics | include packets, Description: Monitor QoS statistics and adjust network traffic patterns as necessary to maintain optimal QoS operation.

Post-Check Verification

```
Command: show qos configuration | include policy-map, Expected Output:
Policy-map name: PE-traffic-policy, Description: Verify the QoS policy
configuration on WHI-PE-14 to ensure it has been properly adjusted.

Command: show interface GigabitEthernet 0/0/0 | include queue,
Expected Output: Queueing strategy: priority-queue, Description:
Verify the interface queue configuration to ensure it has been
properly adjusted.

Command: show qos statistics | include packets, Expected Output:
Packets transmitted: 1234567, Description: Verify QoS statistics to
ensure optimal traffic transmission.
```

Roll-back Procedures

- Step: Restore original QoS configuration, Command: config qos policy-map PE-traffic-policy restore, Description: Restore the original QoS policy configuration to its previous state.

- Step: Restore original interface queue configuration, Command: config interface GigabitEthernet 0/0/0 queue strategy restore, Description: Restore the original interface queue configuration to its previous state.

- Step: Re-monitor network traffic patterns and QoS statistics, Command: show qos statistics | include packets, Description: Re-monitor QoS statistics to ensure network traffic patterns and QoS operation have been restored to their previous state.